

LAC SAINT-PIERRE, QC, Canada

WMO: 711980

Lat: 46.183N

Lon: 72.917W

Elev: 16

StdP: 101.13

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -24.0	(c) -21.2	(d) -29.6	(e) 0.2	(f) -22.5	(g) -26.6	(h) 0.3	(i) -20.0	(j) 16.4	(k) -2.6	(l) 14.0	(m) -4.2	(n) 4.3	(o) 250	(p) 0.685

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.9	27.9	21.1	26.5	20.5	25.3	19.8	22.8	26.0	22.0	25.0	21.2	24.2	6.0	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.7	16.4	24.8	20.9	15.5	24.1	20.0	14.7	23.3	67.5	26.1	64.4	25.1	61.4	24.3	26.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 13.4	(b) 12.0	(c) 10.7	(e) -26.2	(f) 30.4	(g) 3.2	(h) 1.9	(i) -28.5	(j) 31.7	(k) -30.3	(l) 32.9	(m) -32.1	(n) 33.9	(o) -34.4	(p) 35.3		
(a) 13.4	(b) 12.0	(c) 10.7	(e) -27.5	(f) 23.9	(g) 2.5	(h) 1.0	(i) -29.3	(j) 24.6	(k) -30.7	(l) 25.2	(m) -32.1	(n) 25.7	(o) -33.9	(p) 26.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.5	-9.6	-8.8	-2.9	4.8	12.4	18.1	20.7	19.8	16.2	9.2	2.3	-4.8
	DBStd	11.73	6.90	6.10	5.68	4.45	4.45	3.43	2.89	3.13	3.74	4.08	4.79	5.96
	HDD10.0	2482	609	525	401	163	26	0	0	0	2	66	231	460
	HDD18.3	4525	867	758	659	406	191	46	11	21	84	284	479	718
	CDD10.0	1214	0	0	1	7	99	242	332	304	188	40	1	0
	CDD18.3	214	0	0	0	0	5	38	84	67	20	1	0	0
	CDH23.3	837	0	0	0	1	27	156	366	227	61	0	0	0
	CDH26.7	107	0	0	0	0	3	23	56	21	5	0	0	0
(14) Wind	WSAvg	5.4	5.3	5.1	5.3	5.5	5.3	5.1	4.8	4.9	5.2	5.9	5.9	5.8
(15) Precipitation	PrecAvg	1006	73	55	60	85	83	95	109	93	86	99	84	81
	PrecMax	1226	118	99	106	145	164	184	190	193	188	162	141	143
	PrecMin	768	33	30	25	16	39	44	66	39	55	27	26	22
	PrecStd	121	25	18	20	36	34	37	31	34	31	37	26	35
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.5	4.7	11.4	18.9	26.2	29.1	30.4	28.7	27.2	21.8	15.2	8.0
		MCWB	4.6	2.2	7.0	11.4	17.6	21.4	22.3	22.4	22.2	17.4	12.3	6.7
	2%	DB	3.4	2.9	7.0	15.8	23.3	26.7	28.2	27.0	25.1	19.0	12.0	5.6
		MCWB	2.3	1.3	3.9	9.6	15.6	19.8	21.2	21.0	20.4	15.0	9.5	4.4
	5%	DB	2.0	1.7	5.2	12.7	21.0	25.1	26.7	25.7	23.4	16.7	9.9	4.0
		MCWB	1.1	0.6	2.7	8.0	14.2	18.6	20.9	20.2	19.5	13.5	7.8	2.8
	10%	DB	0.1	0.3	3.8	10.6	18.8	23.4	25.3	24.4	21.9	15.1	8.2	2.5
		MCWB	-0.7	-0.6	1.6	6.3	13.4	17.8	20.3	19.7	18.2	12.3	6.3	1.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.0	3.3	9.3	14.2	20.2	22.6	24.3	23.9	22.9	18.0	13.0	6.9
		MCDB	5.0	4.0	10.8	18.6	24.6	27.2	27.9	27.2	25.7	20.9	14.4	7.6
	2%	WB	2.5	1.7	4.8	10.9	18.3	21.4	23.0	22.4	21.6	15.9	10.2	4.6
		MCDB	3.2	2.5	7.0	14.5	21.7	25.2	26.2	25.2	24.1	17.9	11.6	5.5
	5%	WB	1.1	0.8	3.2	8.9	16.2	20.3	22.2	21.6	20.3	14.2	8.3	3.0
		MCDB	1.9	1.6	4.9	12.4	19.6	23.5	25.3	24.3	22.9	16.4	9.9	3.8
	10%	WB	-0.7	-0.4	2.2	7.3	14.5	19.2	21.3	20.7	18.9	12.9	6.8	1.6
		MCDB	0.1	0.7	3.7	10.2	17.7	22.1	24.4	23.5	21.5	15.0	8.5	2.4
(35) Mean Daily Temperature Range	5% DB	MDBR	7.4	8.1	7.1	6.8	7.5	7.3	6.9	6.9	7.0	5.9	5.1	5.4
		MCDBR	8.3	7.3	7.3	10.3	10.5	9.2	8.2	7.9	8.3	8.2	7.5	5.6
		MCWBR	7.7	7.0	5.5	6.7	5.5	4.7	4.4	4.5	5.5	5.6	6.4	5.6
	5% WB	MCDBR	7.5	7.2	7.0	9.6	9.0	7.7	7.0	6.8	8.0	7.5	7.1	5.3
		MCWBR	7.6	7.1	5.7	6.8	6.3	4.9	4.3	4.6	5.6	5.7	6.6	5.6
(40) Clear-Sky Solar Irradiance	taub		0.257	0.285	0.318	0.355	0.382	0.419	0.421	0.410	0.376	0.335	0.308	0.281
	taud		2.336	2.242	2.281	2.359	2.354	2.250	2.254	2.288	2.371	2.492	2.478	2.335
	Ebn at Noon		854	889	907	902	884	849	842	836	836	828	787	773
	Edh at Noon		74	100	112	115	120	134	132	122	103	79	66	67
(44) All-Sky Solar Radiation	RadAvg		1.43	2.33	3.62	4.46	5.13	5.62	5.57	4.89	3.84	2.24	1.34	0.97
	RadStd		0.19	0.26	0.32	0.46	0.52	0.41	0.48	0.23	0.31	0.18	0.17	0.08

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (43 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page